

RF Spectrogram Anomaly Detection with Quantum Kitchen Sinks: Architecture, Representation, and Hardware Validation

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Abstract—The broadcast nature of wireless channels exposes radio-frequency (RF) networks to anomalous and malicious transmissions, making anomaly detection a fundamental requirement for secure spectrum management. Quantum Kitchen Sinks (QKS) offer a lightweight hybrid quantum feature map suitable for near-term quantum devices, yet their behavior on structured signal data remains poorly understood. In this paper, we extend the standard QKS template with multi-depth data re-uploading and ring entanglement, and evaluate the resulting pipeline on controlled RF spectrogram anomaly detection. We introduce a validation-locked five-stage ablation protocol that systematically separates the effects of shallow architecture, re-uploading depth, episode budget, input representation, and classical readout. Across the completed benchmark, Discrete Cosine Transform (DCT) representations consistently dominate raw and Principal Component Analysis (PCA) inputs, moderate-depth entangled QKS configurations form the strongest operating regime, and QKS improves over matched classical direct-readout baselines across all evaluated representation-readout pairs on the held-out test set, with the best configuration reaching a test Area Under the Receiver Operating Characteristic curve (AUROC) of 0.8778 and a test F1 of 0.7995. The study bridges two levels of realism: real measured sub-6 GHz cellular signals on the data side and real-device validation on the *ibm_quebec* Quantum Processing Unit (QPU) on the computing side, with AUROC deviations below 0.013 relative to simulation. These results provide a practical, reproducible framework for deploying QKS-based anomaly detection in wireless networks.

Index Terms—Quantum machine learning, Quantum Kitchen Sinks, anomaly detection, RF spectrograms, hybrid quantum-classical models

I. INTRODUCTION

Spectrum sharing is a fundamental concept in cognitive radio systems, enabling multiple heterogeneous networks to coexist while efficiently utilizing shared radio-frequency (RF) spectrum resources that are inherently limited [1]. This paradigm relies on the open and dynamic nature of the radio spectrum, where wireless transmission propagates over inherently shared channels [2]. However, this openness inherently exposes wireless environments to a wide range of disruptions, including both unintentional disturbances and deliberate attacks such as unauthorized transmissions and malicious interference [3]. Common

examples of these adversarial actions include jamming, sniffing, spoofing, and the injection of anomalous signals [4]. Among these threats, jamming attacks are especially harmful, as adversaries intentionally transmit disruptive signals to interfere with legitimate communications and distort observed spectrum activity [5]. Such interference reduces the signal-to-interference-plus-noise ratio (SINR), resulting in higher error rates, degraded communication quality, and possible system blockage [6]. Furthermore, the rapid advancement of software-defined radio technologies has significantly lowered the barrier to accessing and deploying jamming devices, making these threats more widespread and accessible [7]. Consequently, both civilian and military wireless systems require robust and reliable spectrum monitoring and anomaly detection mechanisms to ensure secure and resilient operation in the presence of such adversarial activities.

Anomaly detection techniques aim to identify both known and unknown interference by detecting deviations from normal signal behavior [8]. Early conventional methods, such as energy detection, cyclostationarity, and matched filtering [9], rely heavily on expert knowledge. Recent works leverage machine learning (ML) on spectrogram datasets for jamming detection [10], using supervised approaches with labeled data [11], [12] and unsupervised methods such as autoencoders and prediction-based models [13], [14], deep Convolutional Neural Networks (CNN)[15], as well as Generative Adversarial Network (GAN)-based frameworks [16]. However, these techniques often suffer from high computational complexity, slow convergence, and limited adaptability in dynamic environments.

Quantum Machine Learning (QML) has emerged at the interface of quantum computing and classical ML, with the goal of exploring whether quantum information processing can enhance the efficiency or performance of learning models [17]. Building on this perspective, QML has also attracted growing interest for anomaly detection in wireless systems, where high-dimensional signal representations and complex interference patterns make efficient feature extraction particularly important [18]. In particular, Quantum Kitchen Sinks (QKS) adapt the classical random kitchen sinks framework of explicit randomized feature

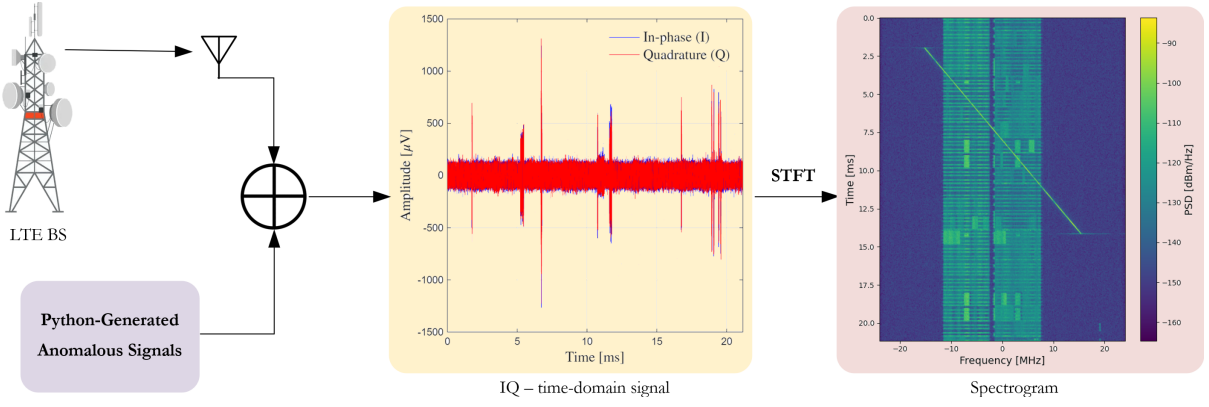


Fig. 1: Diagram of the data acquisition setup. It depicts the measurement of LTE signals, with simulated anomalous signals introduced into the environment. Subsequently, the measured IQ time-domain signals are transformed into spectrograms.

maps to quantum circuits, thereby producing finite-dimensional quantum features that can be processed by lightweight classical readouts [19]. QKS is especially attractive for near-term noisy intermediate-scale quantum (NISQ) devices because the quantum circuit acts as a randomized feature generator rather than a fully trainable variational model, while the optimization burden is shifted to a simple classical head [19], [20]. This yields a modular hybrid pipeline that is straightforward to train, compatible with shallow circuit implementations, and particularly well suited to controlled ablations across both input representation and circuit-design choices. What remains missing in many application-oriented studies, however, is clear characterization of which components of such a hybrid QKS pipeline matter most when the inputs are structured signal representations such as RF spectrograms.

While QML holds significant potential to advance anomaly detection in RF signals, its application to wireless spectrogram analysis remains largely unexplored [21]. To the best of our knowledge, QKS has not yet been applied to anomaly detection in the wireless spectrum, nor validated on real quantum hardware for this task. Motivated by this gap, this paper presents an end-to-end QKS benchmark for controlled RF spectrogram anomaly detection. We extend the standard shallow QKS template [19] with multi-depth data re-uploading and ring entanglement, and evaluate the resulting pipeline through a validation-locked, leakage-free, five-stage protocol that isolates shallow architecture, re-uploading depth, episode budget, input representation, and classical readout effects in sequence, without touching the test split until the final stage. Our study bridges two levels of realism: real measured wireless signals on the data side and real quantum hardware execution on the computing side, while anomaly generation remains deliberately synthetic and controlled.

The contributions of this paper are as follows.

- We construct a labeled spectrogram dataset combining real sub-6 GHz cellular signals with synthetically generated anomalous signals spanning three interference types.
- We extend the standard QKS template with multi-depth data re-uploading and ring entanglement, preserving a

lightweight classical readout stage.

- We introduce a validation-locked, leakage-free five-stage ablation protocol that separates shallow architecture, depth, episode budget, input representation, and readout effects without touching the test split until the final stage.
- We show that the Discrete Cosine Transform (DCT) representations unlock strong QKS performance, with multi-depth QKS consistently improving over matched baselines across all evaluated readouts on the held-out test set.
- Our study bridges two levels of realism by relying on real measured data and by validating the pipeline on the *ibm_quebec* Quantum Processing Unit (QPU), showing deviations in the Area Under the Receiver Operating Characteristic curve (AUROC) below 0.013 relative to simulation.

The remainder of this paper is organized as follows. Section II describes the generation of the spectrogram dataset used for both simulations and experiments. Section III introduces the proposed QKS architecture for anomaly classification. Section IV presents the performance evaluation of the proposed model. Section V discusses the experimental results. Finally, Section VI concludes the paper.

II. SPECTROGRAM DATASET

This study leverages the dataset introduced in [22], which contains raw in-phase and quadrature (IQ) samples measured directly from real Long-Term Evolution (LTE) transmissions across multiple LTE frequency bands. These real-world measurements serve as the primary user signals in cognitive radio networks, providing an authentic foundation for evaluating anomaly detection. To simulate adversarial conditions, anomalous signals are synthetically generated using Python-based simulations and superimposed onto the measured LTE data (see Fig.1). The analyzed sub-6 GHz bands are actively shared between LTE and 5G New Radio (NR) through dynamic spectrum sharing, making the detection of anomalous transmissions directly relevant to today's heterogeneous cellular networks.

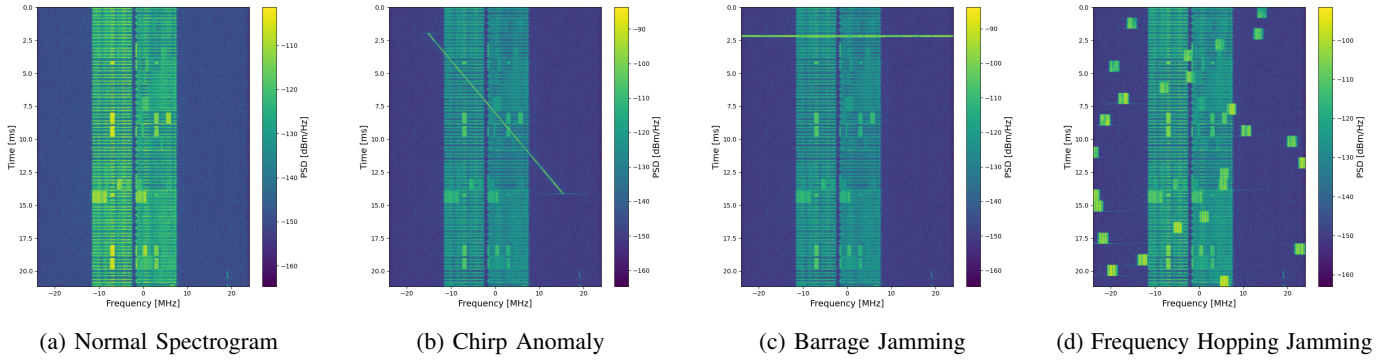


Fig. 2: Example RF spectrograms from the dataset used in this study: (a) normal spectrogram; (b)–(d) spectrograms illustrating three different types of anomalies applied to the signal in (a).

In this configuration, each data sample along the time axis corresponds to a duration of 21.15 ms, while the frequency axis spans a bandwidth of 48 MHz, as detailed in Table I. To emulate anomalous behavior within LTE bands, three types of interference signals are synthetically generated and embedded into the dataset. Specifically, the chirp signal represents an unauthorized transmission [23], while the other two—barrage jamming and frequency-hopping noise jamming (a time-varying form of partial-band interference)—correspond to intentional jamming activities [24]. The temporal duration of these signals is randomly selected within 21.15 ms, while their frequency locations are randomly distributed within the 48 MHz band to ensure diversity. The workflow of the dataset is illustrated in Fig. 1. The embedding process is controlled by the Jamming-to-Signal Ratio (JSR), which quantifies the relative power of the injected anomalies with respect to the LTE signal. Specifically, JSR values ranging from -10 dB to 5 dB, with increments of 2 dB, are considered to model varying interference conditions. The training set consists of 10,800 unique LTE signals, and the test set consists of a disjoint set of 4,062 signals. Since each signal is included in both its normal form and its anomaly-injected form, this yields the 21,600 training samples and 8,124 test samples reported in Table I.

By transforming raw IQ samples into spectrograms, both legitimate transmissions and anomalous activities can be distinguished through their temporal and spectral characteristics. Specifically, a time-domain signal of length 1,300,000 IQ points is formed by combining the original LTE signal with the injected anomalies. Subsequently, we generate the spectrograms by applying a Short-Time Fourier Transform (STFT) using a Hann window, with STFT parameters of 3250 and 8192, and an overlap ratio of 25% [25]. The resulting time–frequency representation is subsequently downsampled to a resolution of 400×400 . Instead of applying a conventional Fast Fourier Transform (FFT) over the entire one-dimensional IQ sequence, the STFT partitions the signal into overlapping segments and computes the Fourier transform on each windowed portion. These localized spectra are then concatenated along the temporal dimension to form a comprehensive time–frequency representation. This approach facilitates the extraction of dis-

tinctive signatures associated with chirp, barrage, and frequency hopping patterns, enabling their differentiation from standard LTE activity. The generated spectrograms are subsequently utilized as inputs to the QKS pipeline for anomaly detection.

TABLE I: Dataset Summary

Train Data samples	21,600
Test Data samples	8,124
Sampling Frequency	61.44 MHz
Number of IQ points per Data sample	1,300,000
Time Duration	21.15 ms
Bandwidth	48 MHz
Spectrogram Size	400×400
Anomaly Signal Classes	Chirp Barrage Jamming Frequency Hopping Jamming
JSR Values [dB]	$[-10, -8, -6, -4, -2, 0, 2, 5]$

An example of the generated dataset is illustrated in Fig. 2. The spectrogram of the normal LTE signal, shown in Fig. 2a, exhibits a structured and relatively stable pattern, where the signal energy remains confined within specific frequency bands corresponding to legitimate transmissions. In contrast, the anomalous cases introduce noticeable distortions in the time–frequency representation. As depicted in Fig. 2b, the chirp signal manifests as a continuous diagonal trace, representing an unauthorized transmission with its time-varying frequency. In Fig. 2c, the barrage jamming spectrogram displays a wideband and dense energy distribution spanning the 48 MHz, indicating simultaneous interference across the entire bandwidth. Meanwhile, Fig. 2d illustrates the frequency-hopping noise jamming, which is characterized by intermittent bursts that rapidly shift across several subbands, resulting in a scattered and discontinuous pattern.

III. QKS PIPELINE AND ABLATION PROTOCOL

A. Input representations

Let $x \in \mathbb{R}^{400 \times 400}$ denote an RF spectrogram. The QKS pipeline does not operate directly on this matrix, but rather on

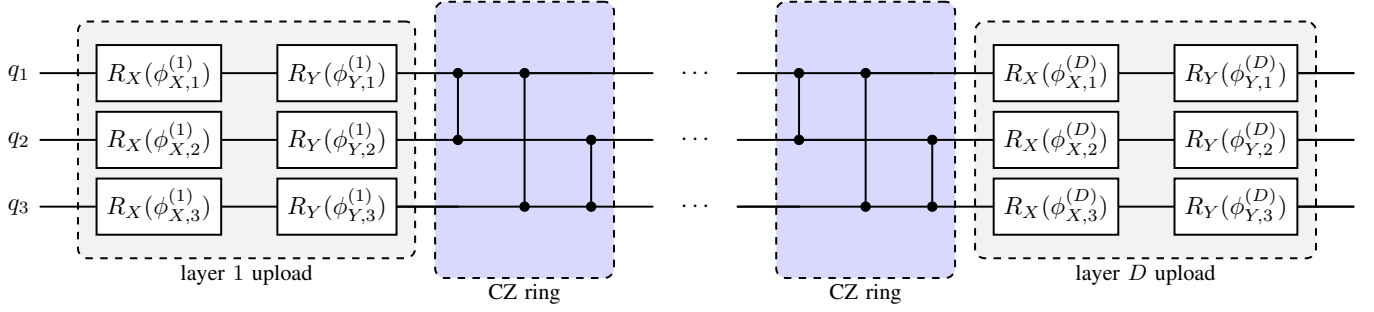


Fig. 3: Illustrative 3-qubit multi-depth QKS episode schematic showing only the first and final layer-specific R_X -then- R_Y uploads. CZ-ring entanglers are inserted only between successive upload layers, so the displayed pattern is layer 1 upload $\rightarrow$ CZ ring $\rightarrow \dots \rightarrow$ CZ ring $\rightarrow$ layer D upload. The special case $D = 1$ recovers the shallow template.

a vector representation $r(x) \in \mathbb{R}^d$ constructed from it. In this study, we consider three such representation families:

$$r_{\text{raw}}(x) = \text{vec}(x), \quad (1)$$

$$r_{\text{DCT}}(x) = \text{vec}(C_{1:k_f, 1:k_t}(x)), \quad (2)$$

$$r_{\text{PCA}}(x) = P_m \text{vec}(x). \quad (3)$$

Here, $\text{vec}(\cdot)$ flattens its matrix argument into a vector, $C_{1:k_f, 1:k_t}(x)$ denotes the upper-left $k_f \times k_t$ coefficient block of the 2-D DCT of x , and P_m denotes the projection onto the first m principal directions learned from the training set.

The raw representation retains the full 400×400 spectrogram and flattens it into a vector in $\mathbb{R}^{160000}$. The DCT family first applies a separable 2-D DCT (type II, with orthonormal normalization) [26] to the spectrogram, retains the upper-left $k_f \times k_t$ block of DCT coefficients, normalizes that block using training-derived statistics, and then flattens it. This retained block corresponds to low-index DCT coefficients, that is, coefficients capturing coarse, slowly varying structure over the 2-D spectrogram array along both the frequency-bin axis and the time-bin axis, rather than low RF frequencies in the original signal.

In the benchmark, we use $k_f \times k_t \in \{16 \times 16, 64 \times 64, 128 \times 128, 256 \times 256\}$. The Principal Component Analysis (PCA) family instead first flattens the spectrogram and then projects it onto an m -dimensional linear subspace, with $m \in \{32, 128, 256, 512, 2048\}$. These representation families compress the same signal in qualitatively different ways. DCT is a fixed transform that explicitly preserves coarse 2-D spectrogram structure, while PCA is a learned linear projection applied to flattened spectrograms and preserves high-variance directions without maintaining the original grid structure in the same explicit way. This distinction is central to the ablation: raw and DCT inputs become vectors only after flattening, whereas PCA already returns a vector representation.

B. Multi-depth QKS featurizer

Given a representation vector $r(x) \in \mathbb{R}^d$, the QKS featurizer constructs one quantum feature block per episode. In line with the original QKS, multiple episodes are used to map the same

input to different randomized quantum states, and each such per-episode state is measured through a fixed observable family to produce a corresponding per-episode feature block. The shallow QKS template [19] corresponds to a single data-upload layer. Here, we extend that template to a depth- D feature map with repeated data re-uploading.

For each episode $e \in \{1, \dots, E\}$ and layer $\ell \in \{1, \dots, D\}$, the model samples two affine maps from the representation space to the n input-dependent circuit angles,

$$\phi_X^{(e,\ell)}(x) = W_X^{(e,\ell)} r(x) + b_X^{(e,\ell)}, \quad (4)$$

$$\phi_Y^{(e,\ell)}(x) = W_Y^{(e,\ell)} r(x) + b_Y^{(e,\ell)}, \quad (5)$$

with $W_X^{(e,\ell)}, W_Y^{(e,\ell)} \in \mathbb{R}^{n \times d}$ and $b_X^{(e,\ell)}, b_Y^{(e,\ell)} \in \mathbb{R}^n$. Equivalently, for each qubit $j \in \{0, \dots, n-1\}$,

$$\phi_{X,j}^{(e,\ell)}(x) = \langle w_{X,j}^{(e,\ell)}, r(x) \rangle + b_{X,j}^{(e,\ell)}, \quad (6)$$

$$\phi_{Y,j}^{(e,\ell)}(x) = \langle w_{Y,j}^{(e,\ell)}, r(x) \rangle + b_{Y,j}^{(e,\ell)}. \quad (7)$$

In the completed benchmark, every entry of $W_X^{(e,\ell)}$ and $W_Y^{(e,\ell)}$ is sampled i.i.d. from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 2$, which was found empirically to yield good results, while every entry of $b_X^{(e,\ell)}$ and $b_Y^{(e,\ell)}$ is sampled i.i.d. from $\mathcal{U}([0, 2\pi])$.

These angle vectors parameterize a layered circuit template. We define the single-layer data-upload operator as

$$U_{e,\ell}(x) = \prod_{j=0}^{n-1} R_Y(\phi_{Y,j}^{(e,\ell)}(x)) R_X(\phi_{X,j}^{(e,\ell)}(x)), \quad (8)$$

and we denote by V_{ring} either the identity (when entanglement is disabled) or the fixed controlled-Z (CZ) ring $(0, 1), (1, 2), \dots, (n-2, n-1), (n-1, 0)$ otherwise. For episode e , the input state $|0\rangle^{\otimes n}$ is mapped to

$$|\psi_e(x)\rangle = U_e(x) |0\rangle^{\otimes n}, \quad (9)$$

where

$$U_e(x) = U_{e,D}(x) V_{\text{ring}} \cdots U_{e,2}(x) V_{\text{ring}} U_{e,1}(x). \quad (10)$$

In other words, each episode re-uploads the same classical representation of the input across multiple randomized layers [27], [28], while entanglement, when enabled, is inserted only

between consecutive upload layers. This preserves a lightweight per-layer circuit structure while allowing the effective quantum feature map to become richer as depth increases. In the circuit schematic below, each upload block is drawn left-to-right as an R_X gate followed by an R_Y gate, which corresponds to the operator product $R_Y R_X$ written above. The special case $D = 1$ recovers the standard shallow QKS template [19]. Figure 3 summarizes this multi-depth circuit structure.

In our method, the measured observable family is taken to consist of all one- and two-body Pauli observables,

$$\mathcal{O}_n = \left\{ \sigma_a^{(i)} : 0 \leq i < n, a \in \{X, Y, Z\} \right\} \cup \left\{ \sigma_a^{(i)} \sigma_b^{(j)} : 0 \leq i < j < n, a, b \in \{X, Y, Z\} \right\}, \quad (11)$$

so that $|\mathcal{O}_n| = 9\binom{n}{2} + 3n$, where the factor 9 comes from the 3×3 possible choices of $(a, b) \in \{X, Y, Z\}$ for each qubit pair (i, j) . This observable family captures both single-qubit statistics and pairwise correlations induced by the feature map, while remaining polynomial in n . The per-episode feature block is therefore

$$\xi_e(x) = (\langle \psi_e(x) | O | \psi_e(x) \rangle)_{O \in \mathcal{O}_n} \in \mathbb{R}^{|\mathcal{O}_n|}, \quad (12)$$

and the final QKS feature vector is obtained by concatenating all per-episode feature blocks as follows,

$$\xi(x) = [\xi_1(x)^\top, \dots, \xi_E(x)^\top]^\top \in \mathbb{R}^{E|\mathcal{O}_n|}. \quad (13)$$

The number of episodes E controls how many independently sampled feature blocks contribute to $\xi(x)$, while the depth D controls how many data re-uploading and entanglement layers are applied within each episode.

C. Five-stage protocol

The benchmark is designed to disentangle the effects of architecture, depth, depth–episode allocation, input representation, and final readout, while keeping the raw test split untouched until the final evaluation. Stages 1–4 operate only on a fixed model-selection subset drawn from the raw training split together with a disjoint validation subset. Stage 5 retrain the finalists on the full raw training split and evaluates them once on the untouched raw test split. The completed benchmark therefore contains $60 + 24 + 4 + 20 + 30 = 138$ evaluations. For the benchmark reported here, the model-selection subset used in Stages 1–4 comprises approximately 20% of the full training split, obtained by stratified random sampling while preserving coverage across abnormal groups; only Stage 5 uses the full training split. This reduced subset keeps the search tractable without collapsing anomaly diversity.

a) Readouts: During Stages 1–4, we deliberately use a single fast linear Support Vector Machine (SVM) readout so that model selection remains focused on the quality of the quantum feature map itself, rather than being confounded by a broader comparison of downstream classifiers. In Stage 5, we then compare alternative corresponding readout families, as they probe different aspects of the same QKS feature geometry, and evaluate their performance under a matched

setting. A linear SVM tests large-margin linear separability [29], logistic regression tests whether the same feature geometry supports a stable probabilistic linear decision rule [30], and the Random Fourier Features (RFF) and Nyström lifts test whether additional gains appear only after a controlled approximate-kernel nonlinearity is applied [31], [32]. This distinction tells us whether the QKS features are already useful under simple linear models or whether their benefit emerges only after a richer nonlinear readout is added. The shortlisted representations are therefore evaluated with five matched readout families. For any Stage 5 readout ρ , let $f_\rho : \mathbb{R}^{E|\mathcal{O}_n|} \rightarrow \mathbb{R}$ denotes its score on $\xi(x)$. We write

$$f_{\text{linSVM}}(\xi) = w_{\text{svm}}^\top \xi + b_{\text{svm}}, \quad (14)$$

$$f_{\text{log}}(\xi) = w_{\text{log}}^\top \xi + b_{\text{log}}, \quad (15)$$

$$f_{\text{RFF-SVM}}(\xi) = w_{\text{rff-svm}}^\top \varphi_{\text{RFF}}(\xi) + b_{\text{rff-svm}}, \quad (16)$$

$$f_{\text{RFF-log}}(\xi) = w_{\text{rff-log}}^\top \varphi_{\text{RFF}}(\xi) + b_{\text{rff-log}}, \quad (17)$$

$$f_{\text{Nys-log}}(\xi) = w_{\text{nys}}^\top \varphi_{\text{Nys}}(\xi) + b_{\text{nys}}. \quad (18)$$

Here, φ_{RFF} and φ_{Nys} denote fixed approximate-kernel feature lifts applied to $\xi(x)$. For the logistic readouts, the corresponding positive-class probability is obtained by applying the sigmoid map to the score. For the approximate-kernel probes, we use matched feature lifts $\varphi_{\text{RFF}}, \varphi_{\text{Nys}} : \mathbb{R}^{E|\mathcal{O}_n|} \rightarrow \mathbb{R}^m$ with projection dimension $m = 1024$, so these heads test whether QKS gains persist after a controlled nonlinear lift of fixed size before the final linear score layer. In the final paired benchmark, we compare each QKS model against a matched direct-readout baseline, meaning that the same shortlisted representation vector $r(x)$ is passed directly to the same readout family instead of first being mapped to $\xi(x)$. The direct-readout baseline and the QKS model therefore share the same data partitions, the same input representation, and the same readout family; only the feature map changes.

b) Protocol: Stages 1–4 rank candidates by validation AUROC first and validation F_1 second. Stage 5 reports test AUROC and test F_1 for the finalists, with F_1 computed from thresholded predictions at a fixed threshold of 0.5.

- 1) **Stage 1: shallow architecture search.** Fix the representation to the 64×64 DCT representation, set depth to $D = 1$, and sweep $n \in \{2, 4, 6, 8, 10\}$ qubits, $E \in \{8, 16, 32, 64, 128, 256\}$ episodes, and entanglement on/off. This stage identifies promising shallow operating points before any additional re-uploading depth is introduced.
- 2) **Stage 2: depth sweep.** Take the top four Stage 1 configurations and sweep depth $D \in \{1, 2, 4, 6, 8, 10\}$ while keeping each candidate’s Stage 1 qubit count, episode count, entanglement setting, and the 64×64 DCT representation fixed. This stage asks whether additional re-uploading depth improves performance once a strong shallow configuration has already been identified.
- 3) **Stage 3: matched depth–episode trade-off.** Retain only the single best Stage 2 family, meaning the same representation, qubit count, and entanglement

setting. Stage 3 does not keep the Stage 2 winner’s exact depth and episode count. Instead, it replaces that winning (D, E) pair with the matched alternatives $(D, E) \in \{(1, 256), (2, 128), (4, 64), (8, 32)\}$. This comparison therefore tests whether the observed Stage 2 gain is attributable to depth itself or instead to a different allocation between circuit depth and the number of episodes.

- 4) **Stage 4: representation comparison.** Evaluate the two strongest Stage 3 candidates across all ten input representations: raw input, four DCT variants, and five PCA variants, after which the best QKS configuration is retained for each shortlisted representation. This stage compares representation families under QKS settings already identified earlier, rather than reopening the full architecture search separately for every representation.
- 5) **Stage 5: final full dataset benchmark.** For the final shortlisted representations, compare matched direct-readout baselines and QKS models across the same five readout families on the full raw training split, and report paired results on the held-out test data.

This design keeps the combinatorial search tractable while yielding interpretable answers to five distinct questions: which shallow architectures are promising, whether depth helps on a fixed shallow shortlist, whether the Stage 2 gain persists under the matched depth–episode trade-off, which input representations remain most effective under the shortlisted QKS settings, and whether those gains persist under a final paired benchmark against matched direct-readout baselines.

IV. RESULTS

A. Ablation study of pipeline design choices

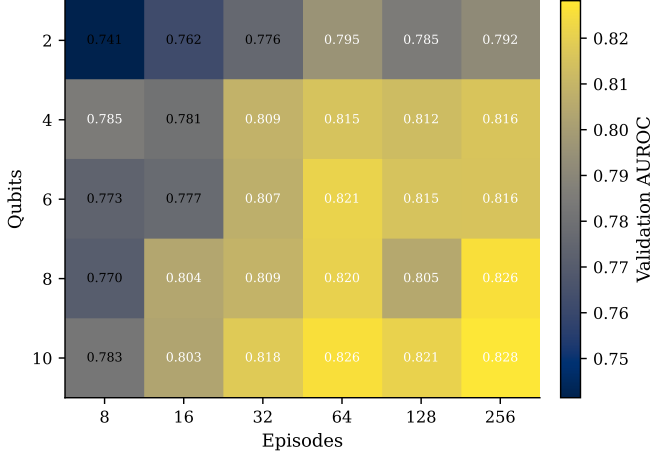
Figure 4 summarizes the evidence from the model-selection stages. Its four panels isolate the entangled Stage 1 sweep (Fig. 4a), the unentangled Stage 1 sweep (Fig. 4b), the Stage 2 depth-response curves (Fig. 4c), and the Stage 3 matched depth–episode trade-off (Fig. 4d).

a) Stage 1: shallow architecture search: On the fixed 64×64 DCT representation, the best shallow model uses 10 qubits, 256 episodes, depth 1, and entanglement, reaching a validation AUROC of 0.828 and a validation F_1 score of 0.749. The second largest alternative is a 8-qubit, 64-episode unentangled configuration, which reaches 0.826 AUROC and 0.729 F_1 . Because the Stage 1 ranking rule uses validation AUROC first and validation F_1 second, the entangled 4-qubit, 64-episode model remains the Stage 1 winner despite that slightly higher F_1 . Figures 4a and 4b therefore suggest that the performance landscape is governed less by a broad monotonic dependence on the number of qubits or QKS episodes than by a limited set of favorable hyperparameter combinations. In the entangled case, the best-performing configurations remain concentrated within a relatively compact region of the sweep, whereas the unentangled case exhibits a flatter and less structured profile, with improvements appearing more sporadically. This suggests that, for the fixed 64×64 DCT input, increasing the size of

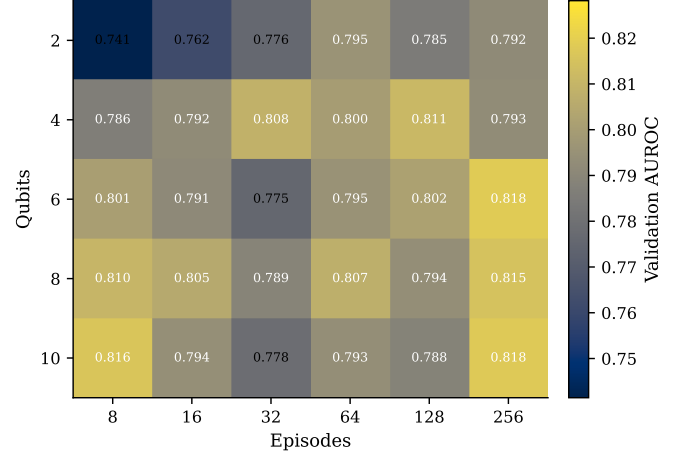
the randomized feature map does not necessarily translate into better performance. Instead, moderate configurations appear sufficient to capture much of the discriminative structure, while larger settings may introduce redundancy and additional model-selection variance.

b) Stage 2: depth sweep: Depth materially improves performance once a strong shallow shell has been identified. Here, the four top performing models from Stage 1 are evaluated, and the best Stage 2 model uses 10 qubits, 256 episodes, depth 4, and entanglement, reaching 0.8502 validation AUROC and 0.7718 validation F_1 . On that same shell, moving from depth 1 to depth 4 raises validation AUROC from 0.8282 to 0.8502 and validation F_1 from 0.7494 to 0.7718. Figure 4c shows a non-monotonic depth trend across the four shortlisted shells: each curve improves once some re-uploading is added, but the strongest entangled $q = 10$, $E = 256$ configuration peaks at depth 4 and then declines. The benchmark therefore favors a moderate re-uploading depth rather than the largest tested depth. This behavior is consistent with repeated data re-uploading enriching the nonlinear feature map up to a moderate depth, after which additional layers increasingly randomize or dilute task-relevant structure relative to the fixed linear readout. The fact that the strongest configurations are consistently entangled further suggests that interaction terms between qubits provide useful additional structure beyond independent single-qubit transformations. Overall, the results indicate that the best performance is obtained by combining entanglement with a moderate re-uploading depth, while increasing circuit depth beyond this regime provides no systematic benefit and may degrade generalization.

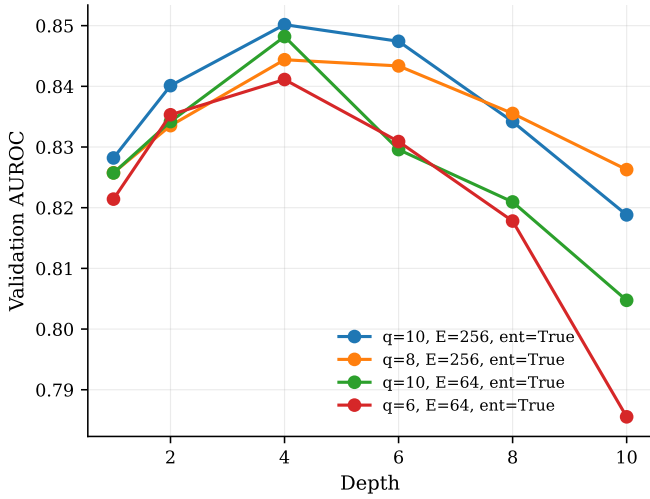
c) Stage 3: matched depth–episode trade-off: Stage 3 introduces a fixed-budget comparison in which the nominal product $D \times E = 256$ is held constant while that budget is re-distributed between circuit depth and the number of episodes. It remains directly anchored to Stage 2 in that it keeps the winning Stage 2 family fixed in terms of representation, qubit count, and entanglement, but it does *not* preserve the Stage 2 winner’s exact $(D, E) = (4, 256)$ setting. Instead, it evaluates the matched alternatives $(D, E) \in \{(1, 256), (2, 128), (4, 64), (8, 32)\}$ in order to test whether the Stage 2 gain is genuinely attributable to depth or whether a similar budget is better spent on additional episodes. The strongest candidate is the 10-qubit, 64-episode, depth-4 model, with 0.8482 validation AUROC and 0.7649 validation F_1 . The 10-qubit, 128-episode, depth-2 alternative remains competitive at 0.8397 AUROC and 0.7531, while the shallow $(D, E) = (1, 256)$ and deep $(D, E) = (8, 32)$ endpoints are weaker. Figure 4d makes the trade-off explicit: within this fixed-allocation comparison, $(D, E) = (4, 64)$ leads on both validation AUROC and validation F_1 , $(D, E) = (2, 128)$ remains the strongest alternative, and the shallow and very deep endpoints trail. The Stage 3 result suggests that the best performance comes from balancing per-episode expressivity and cross-episode diversity: too little depth leaves each episode under-expressive, whereas too few episodes reduce the averaging benefits of the randomized ensemble.



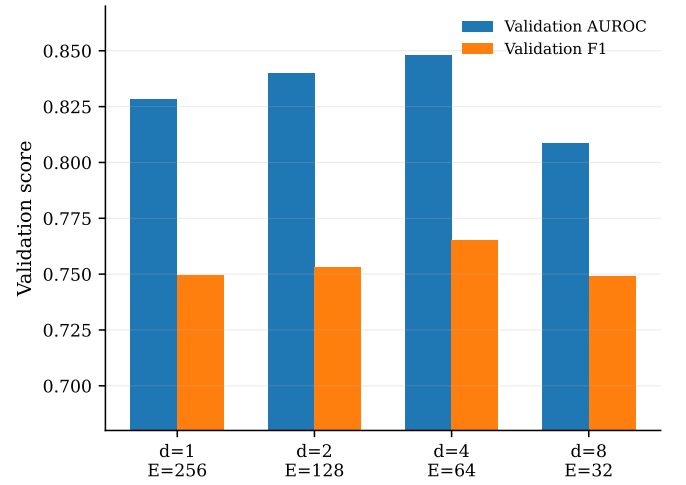
(a) Entangled shallow QKS AUROC heatmap on fixed $\text{dct}64 \times 64$.



(b) Unentangled shallow QKS AUROC heatmap on fixed $\text{dct}64 \times 64$.



(c) Depth-response curves for the four shortlisted shallow configurations.



(d) Validation comparison across matched (D, E) pairs with fixed depth-episode allocation.

Fig. 4: Validation-stage coreset results used to select finalists across Stages 1–3 of the ablation protocol.

d) Stage 4: representation comparison: When the two strongest Stage 3 QKS candidates are compared across all suggested input representations, the DCT family separates sharply from the remaining ones. The three strongest advancing representations are $\text{dct}128 \times 128$, $\text{dct}64 \times 64$, and $\text{dct}256 \times 256$; their best QKS configurations all remain in the entangled 10-qubit regime. The best non-DCT contender, $\text{pca}32$, reaches only 0.5262 validation AUROC, well below the DCT advancing cutoff. Figure 5 makes this pattern explicit across all ten input representations, while Table II reports the three DCT finalists that advance to Stage 5, where the final QKS settings are fixed and for the final evaluation. Under this comparison, the benchmark does not support a representation-agnostic QKS benefit; instead, it identifies a DCT-dominant regime in which structure-preserving classical preprocessing remains part of the effective hybrid model. This sharp DCT advantage is consistent with the idea that anomaly signatures

TABLE II: Stage 4 finalists that advance under the representation comparison and their locked QKS configurations.

Repr.	q	Ep.	D	Ent.	Val AUROC	Val F_1
$\text{dct}128 \times 128$	10	64	4	Y	0.8541	0.7774
$\text{dct}64 \times 64$	10	64	4	Y	0.8482	0.7649
$\text{dct}256 \times 256$	10	128	2	Y	0.8152	0.7299

are organized in the time–frequency plane and are therefore better preserved by low-index DCT coefficients than by raw flattened inputs or PCA directions learned from global variance.

B. Final benchmark on the full dataset

Stage 5 retrains the Stage 4 finalists on the full raw training split and evaluates them once on the held-out raw test split. Among the 15 final representation–readout comparisons, the

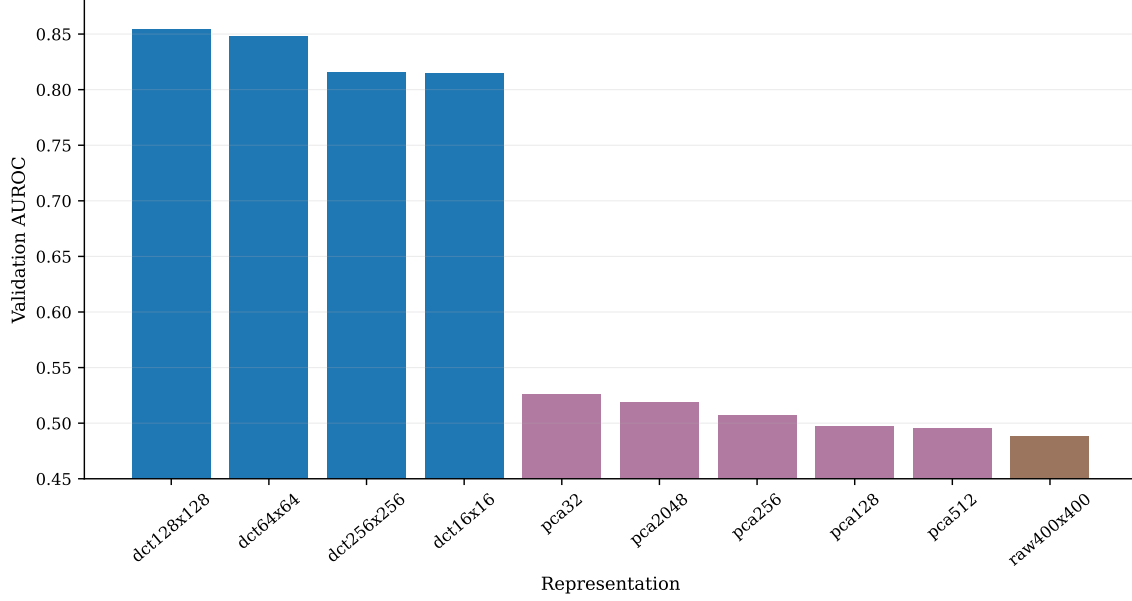


Fig. 5: Stage 4 comparison across input representations. Each bar reports the validation AUROC obtained by applying the two strongest Stage 3 QKS candidates across the ten representation families. DCT representations dominate, while PCA and raw inputs remain well below the advancing cutoff.

highest test AUROC is obtained by `dct64x64 + LinearSvm`, which reaches 0.8778 test AUROC and 0.7938 test F_1 . The highest test F_1 is obtained by `dct64x64 + LinearLogistic`, which reaches 0.7995 at a test AUROC of 0.8716. Figure 6 summarizes the paired direct-readout-versus-QKS test AUROC comparison across all 15 representation-readout combinations, while Table III reports the exact Stage 5 numbers for the six linear cases that exhibit the largest gains. The fact that the best AUROC and best F_1 arise from different DCT configurations suggests that retaining more coefficients is most useful for ranking quality, whereas a more compact representation can yield a cleaner decision boundary at the fixed threshold used for F_1 .

Three key patterns can be identified from these results. First, all 15 rows in Fig. 6 place the QKS pipeline above its matched direct-readout baseline. Here, the direct-readout baseline refers to the same input representation and the same readout family applied directly to the representation vector $r(x)$, without the QKS feature map. The observed AUROC gain is therefore systematic within this finalized benchmark rather than confined to a single isolated representation-readout pair, indicating that the QKS map consistently reshapes the shortlisted DCT inputs into a more discriminative geometry than the matched direct-readout pipeline.

Second, the reporting ΔAUROC and ΔF_1 , which quantify the difference between QKS and direct-readout baseline scores, receptively, show that the largest gains arise for the linear readouts. Across the six linear rows in Table III, the AUROC gains range from +0.0952 to +0.1839, with the largest gap attained by `dct64x64 + LinearSvm`. The approximate-kernel readouts also improve, but their AUROC gains are

visibly smaller. This suggests that most of the benefit is already realized at the level of linear separability, leaving less residual structure for the approximate-kernel heads to recover.

Third, this readout dependence is itself informative about the utility of the QKS features. The fact that the strongest gains arise for the simplest linear readouts indicates that the QKS mapping makes the shortlisted DCT representations more linearly separable on the test split, rather than merely adding redundancy that only richer nonlinear heads can exploit. In practical terms, the QKS map appears to act primarily as a representation enhancer whose effect is strongest when the downstream classifier is deliberately simple, while still yielding a positive, but smaller, lift for the nonlinear readouts.

C. Validation on real quantum hardware

To assess the hardware-realism aspect of the study, we performed a focused validation on quantum hardware. Due to the cost of real-device execution, all experiments were conducted with an entangled feature map using 4 qubits, a single-layer depth, and 32 episodes, with expectation values estimated from 4096 shots per circuit. These experiments were run on the *ibm_quebec* QPU from 25th to 29th March 2026, using the `dct128x128` representation. In contrast to the completed simulation study, the measured observable family was restricted to all single-qubit Z observables and pairwise ZZ observables,

$$\mathcal{O}_{Z,ZZ}^{(n)} = \{\sigma_Z^{(i)}\}_{i=0}^{n-1} \cup \{\sigma_Z^{(i)}\sigma_Z^{(j)}\}_{0 \leq i < j < n}, \quad (19)$$

so that $|\mathcal{O}_{Z,ZZ}^{(n)}| = \frac{n(n+1)}{2}$, instead of the full one- and two-body Pauli family considered in simulation. In addition, the dataset size was reduced to approximately 10% of the

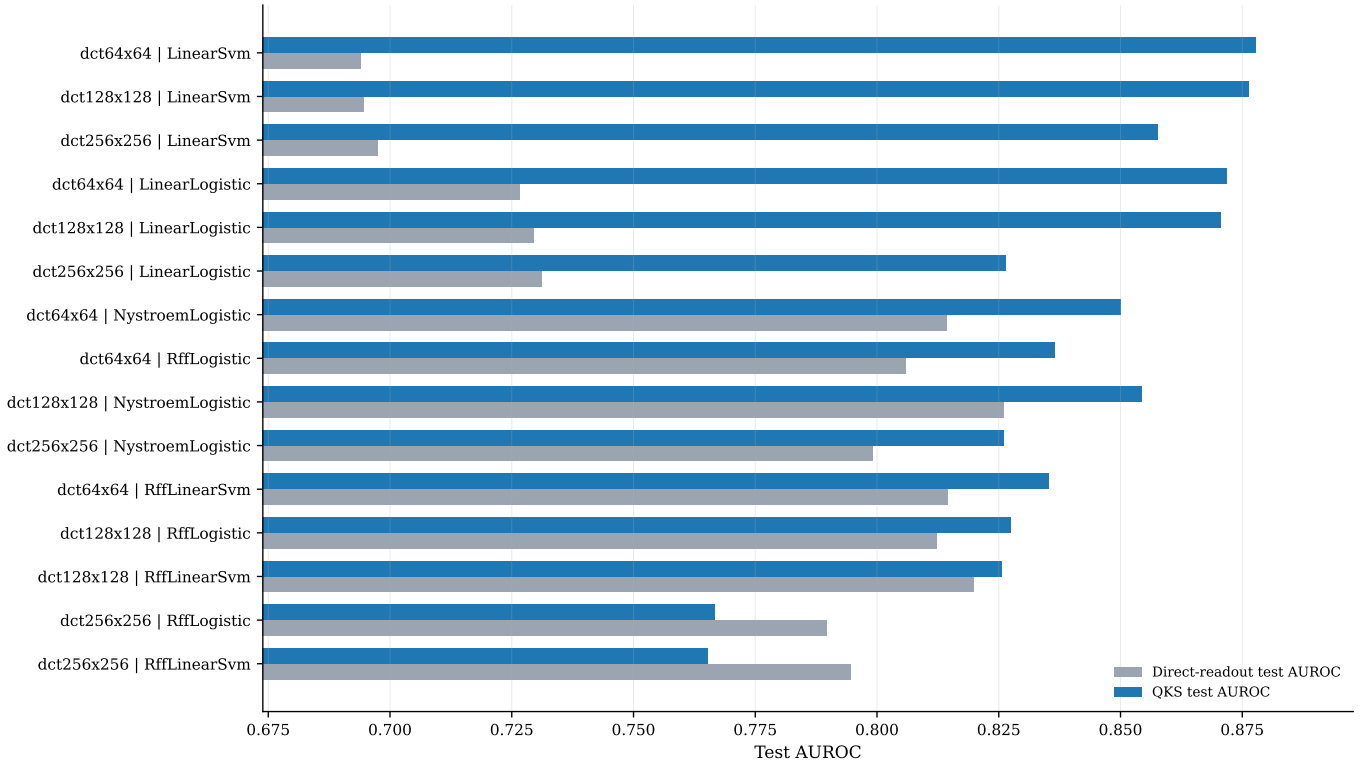


Fig. 6: Stage 5 paired test AUROC comparison between matched direct-readout baselines and QKS pipelines. Each row corresponds to one representation–readout comparison on the untouched raw test split after the locked finalists from a single finalized benchmark run are retrained on the full raw training split. The largest AUROC gaps appear for the linear readouts, while the approximate-kernel readouts also benefit from QKS but by smaller margins.

TABLE III: Compact Stage 5 benchmark slice for the linear readouts.

Repr.	Readout	Direct AUROC	QKS AUROC	Δ AUROC	Direct F_1	QKS F_1	ΔF_1
dct64x64	LinearSvm	0.6939	0.8778	0.1839	0.6515	0.7938	0.1423
dct64x64	LinearLogistic	0.7266	0.8717	0.1451	0.6607	0.7995	0.1389
dct128x128	LinearSvm	0.6945	0.8763	0.1818	0.6519	0.7969	0.1451
dct128x128	LinearLogistic	0.7294	0.8705	0.1411	0.6614	0.7978	0.1363
dct256x256	LinearSvm	0.6975	0.8576	0.1601	0.6558	0.7754	0.1195
dct256x256	LinearLogistic	0.7312	0.8264	0.0952	0.6635	0.7739	0.1104

original training and test sets. The search space was deliberately restricted so as to remain within the available quantum budget and to limit the practical overhead associated with queue times and repeated hardware calls. As a result, this experiment was not intended to reproduce the full simulation study, but rather to evaluate the practical viability of the approach under genuine real-device conditions. It should therefore be viewed as a focused real-QPU benchmark aimed at assessing whether the learned representation retains meaningful predictive structure when executed on an actual superconducting quantum processor.

The resulting comparison between real-QPU and simulator performance is reported in Table IV for several classical readout models, with the simulator baseline obtained by re-running the pipeline under exactly the same configuration used on real-QPU—identical feature map, number of qubits, circuit

depth, number of episodes, observable family $\mathcal{O}_{Z,ZZ}^{(n)}$, and train/test splits—so that the execution backend is the only varying factor. Overall, the real-hardware results remain close to their simulated counterparts, with only small deviations in AUROC across all tested readouts. Here, HS denotes the hardware–simulator difference:

$$\Delta_{\text{HS}}\text{AUROC} = \text{AUROC}_{\text{QPU}} - \text{AUROC}_{\text{sim}}.$$

More specifically, the observed AUROC gaps range from -1.281×10^{-2} to $+7.132 \times 10^{-3}$, indicating that the ranking quality of the QKS features is largely preserved under real-device execution despite the reduced measurement setting. The F1-score differences,

$$\Delta_{\text{HS}}F_1 = F1_{\text{QPU}} - F1_{\text{sim}},$$

Readout	Hardware AUROC	Hardware F1	Simulator AUROC	Simulator F1	Δ_{HS} AUROC	Δ_{HS} F1
LinearSvm	0.738302	0.605144	0.731170	0.657754	$+7.132 \times 10^{-3}$	-5.261×10^{-2}
LinearLogistic	0.720810	0.658851	0.733619	0.645161	-1.281×10^{-2}	$+1.369 \times 10^{-2}$
RffLinearSvm	0.807438	0.719892	0.805798	0.704762	$+1.640 \times 10^{-3}$	$+1.513 \times 10^{-2}$
RffLogistic	0.803232	0.736967	0.805961	0.702247	-2.729×10^{-3}	$+3.472 \times 10^{-2}$
NystroemLogistic	0.813637	0.748166	0.815457	0.716418	-1.819×10^{-3}	$+3.175 \times 10^{-2}$

TABLE IV: Comparison of real-QPU and simulator performance across classical readout models, both evaluated under identical settings: 4-qubit entangled feature map, single-layer depth, 32 episodes, $\mathcal{O}_{Z,ZZ}$ observables, train/test split.

are somewhat more variable, but remain favorable for most readouts. In particular, the `NystroemLogistic` readout achieves the best real-QPU performance, with an AUROC of 0.813637 and an F1 score of 0.748166, compared with 0.815457 and 0.716418 in simulation. Taken together, these results indicate that, within this deliberately reduced experimental setting, the proposed pipeline remains robust to real-device execution and preserves a substantial fraction of the predictive structure observed in simulation.

V. DISCUSSION AND LIMITATIONS

a) Multi-depth QKS acts as a representation enhancer:

The completed benchmark does not suggest that QKS uniformly dominates classical baselines. Instead, it shows that repeated data re-uploading can improve separability in specific representation regimes, with the strongest gains appearing once depth is increased beyond the shallow template but kept below the largest tested values. Entanglement also appears to contribute positively in the strongest regimes: entangled configurations dominate among the top-performing shallow models and outperform their unentangled counterparts in most matched comparisons, although this benefit is not systematic across the entire search space. Taken together, these results suggest that the strongest QKS representations arise from a combination of a suitable input representation, moderate re-uploading depth, and useful inter-qubit interactions, rather than from indiscriminately increasing circuit complexity.

b) *The strongest regime is DCT-specific:* The feature-representation stage is decisive: all top-performing configurations use DCT variants, whereas PCA and raw inputs do not remain competitive. This suggests that, for spectrogram-based RF anomaly detection, performance depends not on the QKS feature map alone, but on its combination with a suitable classical representation. Our interpretation is that DCT is more effective because it preserves the structured organization of anomaly patterns in the time–frequency plane while concentrating informative content into a compact set of coefficients. By contrast, PCA emphasizes directions of maximal global variance, which can mix distant time–frequency regions and obscure features that are more relevant for anomaly discrimination. Raw inputs, in turn, likely preserve excessive irrelevant variability and redundancy, thereby reducing the ability of the QKS feature map to produce a representation aligned with the discriminative structure of the task.

c) *Readout dependence remains informative:* All final readouts benefit from QKS, yet the largest paired gains appear

for the linear probes. This is informative as it shows that QKS is changing the geometry seen by simple classical classifiers, not merely adding redundancy that only richer downstream models can exploit. In practical terms, this is the main value of the QKS features in this study: they act as a representation enhancer whose effect is largest when the downstream classifier is deliberately simple, while still yielding a positive, but smaller, lift for the nonlinear readouts.

d) *Real-device execution preserves feature quality:* The hardware study confirms that QKS features retain meaningful predictive structure under real-device conditions. Although the QPU experiment uses a reduced configuration with fewer qubits, shallower depth, a restricted observable set, and a smaller dataset, the results remain within 0.013 AUROC of their simulated counterparts across all tested readouts. Notably, the ranking order across readouts is preserved between hardware and simulation, with `NystroemLogistic` achieving the best performance in both settings. This consistency indicates that the feature geometry learned by the QKS pipeline is not critically disrupted by hardware noise at this scale, and supports the practical viability of the approach on near-term superconducting processors. We emphasize that this experiment was not designed to reproduce the full simulation benchmark, but to assess whether the pipeline remains functional under genuine real-device constraints, a question it answers affirmatively. Taken together, these observations suggest that the study probes realism along both the data and hardware axes, placing the benchmark closer to realistic hybrid deployment conditions.

e) *Limitations:* The evidence remains task-specific. The dataset is controlled rather than drawn from a live operational RF deployment, and generalization to unseen interference types or real-world spectrum conditions has not been tested. The completed study uses one finalized run rather than a broad seed ensemble, so stability across random initializations of the affine maps has not been quantified. The classical readout family is intentionally restricted to fast probes and kernel approximations, and stronger classical heads may reduce or eliminate the observed QKS gains. We therefore make no universal claim about QKS on every anomaly detection task, and we do not interpret the present results as evidence of quantum advantage.

VI. CONCLUSION

This paper studied Quantum Kitchen Sinks for RF spectrogram anomaly detection under a controlled, leakage-free evaluation protocol. Starting from the standard shallow QKS

template, we introduced a multi-depth variant with repeated data re-uploading and optional ring entanglement, and evaluated it on spectrograms built from real measured sub-6 GHz cellular signals with synthetically injected anomalous transmissions. The resulting benchmark was designed not only to compare models, but also to identify which components of the hybrid pipeline matter most for this task.

The results support a clear picture within this benchmark. QKS does not provide a representation-agnostic improvement over classical baselines; instead, its utility depends strongly on the interaction between the quantum feature map, the input representation, and the downstream readout. The strongest results arise in DCT-based regimes, where moderate-depth entangled QKS configurations outperform their matched direct-readout baselines across all evaluated final readout families on held-out test data, reaching a best test AUROC of 0.8773 and a best test F_1 of 0.8032. The same favorable pattern is not reproduced with raw or PCA-based inputs, and the largest gains do not come from the largest tested models. In addition, the reduced real-device validation on the *ibm_quebec* QPU preserves the main predictive trends seen in simulation, with AUROC deviations below 0.013. Future work should extend the study to additional RF anomaly families and test whether the same operating regimes persist under broader hardware-aware execution constraints.

VII. ACKNOWLEDGEMENT

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REFERENCES

- [1] X. Jiang, P. Li, B. Li, Y. Zou, and R. Wang, "Intelligent Jamming Strategies for Secure Spectrum Sharing Systems," *IEEE Trans. Commun.*, vol. 70, no. 2, pp. 1153–1167, 2022.
- [2] Y. Zou, J. Zhu, X. Wang, and L. Hanzo, "A Survey on Wireless Security: Technical Challenges, Recent Advances, and Future Trends," *Proc. IEEE*, vol. 104, no. 9, pp. 1727–1765, 2016.
- [3] A. S. Ali, W. T. Lunardi, G. Singh, L. Bariah, M. Baddeley, M. A. Lopez, J.-P. Giacalone, and S. Muhaidat, "RF Jamming Dataset: A Wireless Spectral Scan Approach for Malicious Interference Detection," *IEEE Commun. Mag.*, vol. 62, no. 11, pp. 114–120, 2024.
- [4] M. Lichtman, R. P. Jover, M. Labib, R. Rao, V. Marojevic, and J. H. Reed, "LTE/LTE-A Jamming, Spoofing, and Sniffing: Threat Assessment and Mitigation," *IEEE Commun. Mag.*, vol. 54, no. 4, pp. 54–61, 2016.
- [5] M. Lichtman, J. D. Poston, S. Amuru, C. Shahriar, T. C. Clancy, R. M. Buehrer, and J. H. Reed, "A Communications Jamming Taxonomy," *IEEE Secur. Priv.*, vol. 14, no. 1, pp. 47–54, 2016.
- [6] H. Pirayesh and H. Zeng, "Jamming Attacks and Anti-Jamming Strategies in Wireless Networks: A Comprehensive Survey," *IEEE Commun. Surv. Tutor.*, vol. 24, no. 2, pp. 767–809, 2022.
- [7] X. Wang, J. Wang, Y. Xu, J. Chen, L. Jia, X. Liu, and Y. Yang, "Dynamic Spectrum Anti-Jamming Communications: Challenges and Opportunities," *IEEE Commun. Mag.*, vol. 58, no. 2, pp. 79–85, 2020.
- [8] S. Rajasegarar, C. Leckie, and M. Palaniswami, "Anomaly Detection in Wireless Sensor Networks," *IEEE Wireless Commun.*, vol. 15, no. 4, pp. 34–40, 2008.
- [9] T. Yucek and H. Arslan, "A Survey of Spectrum Sensing Algorithms for Cognitive Radio Applications," *IEEE Commun. Surv. Tutor.*, vol. 11, no. 1, pp. 116–130, 2009.
- [10] A. Krause, M. D. Khursheed, P. Schulz, F. Burmeister, and G. Fettweis, "Digital Twin of the Radio Environment: A Novel Approach for Anomaly Detection in Wireless Networks," in *2023 IEEE Globecom Workshops (GC Wkshps)*. IEEE, 2023, pp. 1307–1312.
- [11] Z. Wu, Y. Zhao, Z. Yin, and H. Luo, "Jamming Signals Classification Using Convolutional Neural Network," in *2017 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*. IEEE, 2017, pp. 062–067.
- [12] H. Xu, X. Ma, C. Wang, X. Wang, C. Xu, F. Gao, and L. Kong, "A neural network approach for wireless spectrum anomaly detection in 5G-unlicensed network," *CCF Transactions on Pervasive Computing and Interaction*, vol. 4, no. 4, pp. 465–473, 2022.
- [13] S. Rajendran, W. Meert, V. Lenders, and S. Pollin, "Unsupervised Wireless Spectrum Anomaly Detection With Interpretable Features," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 3, pp. 637–647, 2019.
- [14] N. Tandiya, A. Jauhar, V. Marojevic, and J. H. Reed, "Deep Predictive Coding Neural Network for RF Anomaly Detection in Wireless Networks," in *2018 IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2018, pp. 1–6.
- [15] J. Ren, X. Zhang, and Y. Xin, "Using Deep Convolutional Neural Network to Recognize LTE Uplink Interference," in *2019 IEEE Wireless Communications and Networking Conference (WCNC)*. IEEE, 2019, pp. 1–6.
- [16] X. Zhou, J. Xiong, X. Zhang, X. Liu, and J. Wei, "A Radio Anomaly Detection Algorithm Based on Modified Generative Adversarial Network," *IEEE Wireless Commun. Lett.*, vol. 10, no. 7, pp. 1552–1556, 2021.
- [17] P. Liao, X. Wang, Y. Shan, L. An, and S. Mao, "Wireless Sensing in Artificial Intelligence of Things: A General Quantum Machine Learning Framework," *IEEE Netw.*, vol. 39, no. 3, pp. 207–214, 2025.
- [18] I. Ahmad, L. Yang, Y. Alkhrijah, A. Almadhor, M. A. Alawad, L. Peng, and P.-H. Ho, "Quantum Machine Learning for Anomaly Detection: The Future of Smarter and Safer IoT Networks," *IEEE Netw.*, vol. 40, no. 2, pp. 30–38, 2026.
- [19] C. M. Wilson, J. S. Otterbach, N. Tezak, R. S. Smith, G. E. Crooks, and M. P. da Silva, "Quantum kitchen sinks: An algorithm for machine learning on near-term quantum computers," 2018. [Online]. Available: <https://arxiv.org/abs/1806.08321>
- [20] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, no. 9, pp. 625–644, 2021.
- [21] E. Mohammadisavadkoobi, N. Shafiabady, and J. Vakilian, "A Systematic Review on Quantum Machine Learning Applications in Classification," *IEEE Trans. Artif. Intell.*, vol. 7, no. 1, pp. 4–21, 2026.
- [22] J. Kim, H. Kim, and B. Kim, "Wireless Anomaly Signal Dataset (WASD): An Open Dataset for Wireless Cellular Spectrum Monitoring and Anomaly Detection," *IEEE Access*, vol. 12, pp. 196 240–196 248, 2024.
- [23] J. Kim, S. Oh, H. Kim, and B. Kim, "Spectrum Anomaly Detection Using Deep Neural Networks: A Wireless Signal Perspective," *IEEE Access*, vol. 13, pp. 154 948–154 959, 2025.
- [24] P. Lohan, B. Kantarci, M. A. Ferrag, N. Tihanyi, and Y. Shi, "From 5G to 6G Networks: A Survey on AI-Based Jamming and Interference Detection and Mitigation," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 3920–3974, 2024.
- [25] B. Kim, S.-H. Kong, and S. Kim, "Low Computational Enhancement of STFT-Based Parameter Estimation," *IEEE J. Sel. Top. Signal Process.*, vol. 9, no. 8, pp. 1610–1619, 2015.
- [26] G. Wallace, "The JPEG Still Picture Compression Standard," *IEEE Trans. Consum. Electron.*, vol. 38, no. 1, pp. xviii–xxxiv, 1992.
- [27] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data re-uploading for a universal quantum classifier," *Quantum*, vol. 4, p. 226, 2020.
- [28] M. Schuld, R. Sweke, and J. J. Meyer, "Effect of data encoding on the expressive power of variational quantum-machine-learning models," *Physical Review A*, vol. 103, p. 032430, 2021.
- [29] C. Cortes and V. Vapnik, "Support-Vector Networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [30] D. R. Cox, "The Regression Analysis of Binary Sequences," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 20, no. 2, pp. 215–232, 1958.
- [31] A. Rahimi and B. Recht, "Random Features for Large-Scale Kernel Machines," in *Advances in Neural Information Processing Systems 20*. Curran Associates, Inc., 2007, pp. 1177–1184.
- [32] C. K. I. Williams and M. Seeger, "Using the Nyström Method to Speed Up Kernel Machines," in *Advances in Neural Information Processing Systems 13*. MIT Press, 2001, pp. 682–688.